Interpretable Multi-Modal Reasoning Framework for Protein Stability Prediction: Bridging Deep Learning and Rational Design

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Abstract

Predicting the effects of amino acid mutations on protein stability ($\Delta\Delta G$) is crucial for protein engineering, drug design, and understanding disease mechanisms. While recent deep learning approaches have achieved impressive predictive accuracy, they often lack interpretability and strategic reasoning capabilities essential for practical protein design tasks. We present a novel multi-modal framework that combines Graph Neural Networks (GNNs) for 3D structural information with Transformers for sequence modeling, augmented with multiple interpretable reasoning strategies for mutation selection. Our framework introduces: (1) a cross-modal fusion architecture that integrates spatial and sequential protein features, (2) mutation-aware attention mechanisms that focus on local structural perturbations, (3) five distinct search strategies (greedy, beam search, Monte Carlo, ensemble, and constraint-based) for exploring mutation space, and (4) chain-of-thought reasoning modules that generate human-interpretable explanations for each prediction. Evaluated on benchmark datasets including ProTherm and SKEMPI, our approach achieves competitive accuracy (Pearson $\rho = 0.85$) while providing actionable insights through molecular rationale generation. The framework successfully identifies stabilizing mutations with confidence scores and contextual awareness, demonstrated through case studies in antibody engineering and enzyme thermostability optimization. This work bridges the gap between predictive models and practical protein engineering by providing both accurate predictions and interpretable decision-making processes.

Keywords: Protein engineering, $\Delta\Delta G$ prediction, Graph neural networks, Transformers, Multi-modal learning, Interpretable AI, Rational protein design

1 Introduction

Protein thermodynamic stability, quantified by the change in Gibbs free energy upon mutation ($\Delta\Delta G = G_{\text{mutant}} - G_{\text{wild-type}}$), is a fundamental property that governs protein folding, function, and evolutionary fitness Tokuriki and Tawfik [2009]. Accurate prediction of mutation effects enables rational protein engineering for diverse applications including therapeutic antibody development Jain et al. [2017], industrial enzyme optimization [Turner, 2009], and understanding disease-causing mutations Yue and Moult [2005].

Traditional computational methods employ physics-based energy functions (FoldX Schymkowitz et al. [2005], Rosetta Kellogg et al. [2011]), achieving moderate accuracy (Pearson $\rho \approx 0.5$) but requiring extensive computational resources. Recent machine learning approaches leverage deep neural networks, improving accuracy to $\rho \approx 0.75 - 0.80$ Meier et al. [2021], Cao et al. [2019]. However, these methods face three critical limitations:

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Challenge 1: Interpretability Gap. Black-box models provide predictions without mechanistic explanations, hindering adoption by experimental researchers who need to understand *why* a mutation is predicted to stabilize or destabilize a protein.

Challenge 2: Modal Integration. Proteins exhibit multi-scale organization—sequence determines structure, which determines function. Existing methods process sequence and structure independently, missing critical cross-modal dependencies.

Challenge 3: Strategic Reasoning. Protein design requires systematic exploration of mutation space with multiple competing objectives (stability, solubility, activity). Current approaches evaluate mutations individually without strategic search capabilities.

Our Contributions

We present an **Interpretable Multi-Modal Reasoning Framework** that addresses these challenges through:

1. **Hybrid GNN-Transformer Architecture** (§**??**): Novel cross-modal fusion integrating geometric graph neural networks for 3D structure with transformers for sequence, using mutation-aware attention mechanisms.

2. **Strategic Search Framework** (§**??**): Five complementary strategies (greedy, beam search, Monte Carlo, ensemble, constraint-based) with rigorous mathematical formulation and convergence analysis.

3. **Chain-of-Thought Reasoning** (§**??**): Automated generation of human-interpretable explanations grounded in structural biology principles, with confidence calibration and molecular rationale.

4. **Comprehensive Evaluation** (§**??**): Achieves state-of-the-art performance ($\rho = 0.85$) on ProTherm with superior interpretability (human evaluation: 4.2/5.0), demonstrated through three real-world case studies.

The rest of this paper is organized as follows: Section 2 reviews related work. Section 3 details the prediction model architecture. Section 4 presents the reasoning framework with mathematical formulation. Section 5 describes experiments and results. Section 6 discusses findings and limitations. Section 7 concludes with future directions.

2 Related Work

The progression of computational methods for predicting the change in stability upon mutation ($\Delta\Delta G$) has evolved through three distinct eras, each marked by a trade-off between speed, accuracy, and interpretability. The initial phase focused on **physics-based approaches** such as FoldX Schymkowitz et al. [2005] and Rosetta Kellogg et al. [2011], which employed molecular mechanics force fields and rigorous energy minimization techniques. While these methods offered mechanistic interpretability by directly modeling physical interactions, they were computationally expensive, often requiring minutes per mutation, and yielded only moderate accuracy. The subsequent development of **statistical machine learning (ML) techniques** utilized hand-crafted features coupled with classical algorithms, exemplified by I-Mutant2.0 Capriotti et al. [2005] and the work of Cheng et al. Cheng et al. [2006]. This statistical approach increased speed and incrementally improved accuracy to the range of $\rho \approx 0.6 - 0.7$.

The field achieved its current state-of-the-art accuracy with the advent of **deep learning architectures**. This era bifurcated into two main strategies: **sequence-based deep learning** and **structure-based deep learning**. Sequence-based methods, notably Protein Language Models (PLMs) such as those developed by Meier et al. Meier et al. [2021] and Rives et al. Rives et al. [2021], achieved the highest predictive performance, with correlations reaching $\rho \approx 0.75 - 0.78$ by leveraging unsupervised pre-training on millions of evolutionary sequences. Conversely, **structure-based deep learning** utilized Graph Neural Networks (GNNs) and 3D Convolutional Neural Networks (CNNs), including DeepDDG Cao et al. [2019], Li et al.'s work Li et al. [2020], and Zhou et al.'s model Zhou et al. [2020], to explicitly model the protein's tertiary structure. While these models successfully captured local physical interactions, they typically lagged slightly in accuracy ($\rho \approx 0.70 - 0.73$) due to their reduced ability to incorporate large-scale evolutionary context. Initial **hybrid approaches** attempted to combine these modalities via simple concatenation or late fusion, but they lacked a principled mechanism for truly synergistic cross-modal interaction.

Despite achieving high numerical accuracy, current deep learning paradigms present critical limitations for rational, high-stakes therapeutic design. First, the models primarily rely on post-hoc interpretation methods—such as attention visualization Vig and et al. [2021], saliency maps Sundararajan et al. [2017], and concept-based explanations Kim and Doshi-Velez [2018]—which provide suggestive input correlations rather than integrated, human-understandable mechanistic reasoning. This opacity fundamentally undermines the trust required for clinical translation. Second, existing design frameworks, including those based on genetic algorithms Romero and Arnold [2009], Bayesian optimization Laine and Kaski [2021], and reinforcement learning Angermueller et al. [2019], treat the design process as a heuristic search through a vast sequence space. Critically, these methods fail to integrate the prediction and search steps with explicit, verifiable mechanistic reasoning, thus providing design choices without scientific justification and lacking the strategic search capabilities necessary for truly efficient and accountable protein engineering.

3 Methodology

3.1 Problem Formulation

Input:

- Protein sequence $\mathbf{s} = (s_1, \ldots, s_L)$ where $s_i \in \mathcal{A}$ (20 amino acids)
- 3D structure graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ represents residues with coordinates $\{\mathbf{r}_i\}$
- Mutation $m = (p, s_p^{\text{wt}}, s_p^{\text{mut}})$: position, wild-type, mutant

Output:

- $\Delta\Delta G$ prediction with confidence $c \in [0, 1]$
- Reasoning chain $\mathcal{R} = \{r_1, \ldots, r_k\}$ explaining the prediction
- Molecular rationale connecting prediction to structural/sequence features

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